The Slutsky Equation and Intertemporal Choice

Supplementary Graduate Economics Lecture

30 minutes

1 The Slutsky Equation (5 minutes)

Begin with the familiar expression

$$\boxed{\frac{\partial c_i}{\partial p_j} = \frac{\partial h_i}{\partial p_j} - c_j \frac{\partial c_i}{\partial W}.} \tag{1}$$

Here $c_i(\mathbf{p}, W)$ is *Marshallian demand*, obtained by maximizing utility subject to a budget constraint. It depends upon all prices and money income. (In an inter-temporal framework, W is the present value of the stream of payments. It is also called wealth.) $h_i(\mathbf{p}, \bar{u})$ is *Hicksian demand*, obtained by minimizing expenditure subject to a fixed utility target $\bar{u}$. It depends upon all prices and utility Utility is not observable, but it is a foundational concept for an economist.

In this equation, the first term on the right is the substitution effect; the second is the income effect. Derivatives are evaluated at the same initial optimum, where $h_i = c_i$. The Hicksian demand is a theoretical or philosophical idea, but it must be equal to the Marshallian demand that we can actually see and observe.

2 Elasticity Form (5 minutes)

Multiply the Slutsky equation by p_j/c_i:

$$\frac{p_j}{c_i}\frac{\partial c_i}{\partial p_j} = \frac{p_j}{c_i}\frac{\partial h_i}{\partial p_j} - \frac{p_j c_j}{c_i}\frac{\partial c_i}{\partial W}. \tag{2}$$

Define the Marshallian and Hicksian price elasticities and the wealth elasticity:

$$\varepsilon_{ij}^M = \frac{\partial c_i}{\partial p_j}\frac{p_j}{c_i}, \qquad \varepsilon_{ij}^H = \frac{\partial h_i}{\partial p_j}\frac{p_j}{h_i}, \qquad \eta_i = \frac{\partial c_i}{\partial W}\frac{W}{c_i}.$$

The price elasticity is the percentage change in quantity demanded divided by the percentage change in price. The income elasticity of demand is the percentage change in quantity demanded divided by the percentage change in income. These are numbers without units, and that is why economist usually work with these concepts. Again, the Hicksian price elasticity is a theoretical concept that is foundational for an economist. It captures the curvature of the indifference curve.

One can think of the percentage change in the demand for peanut butter with respect to the percentage change in the price of peanut butter. An one can just as well contemplate the percentage change in the demand for peanut butter with respect to the percentage change in the price of jelly. So there are own-price elasticities and cross-price elasticities.

If the demand for peanut butter goes down when the price of jelly rises, then the two goods are complements. If it increases, then they are substitutes. If the income elasticity for a good is negative, then it is inferior. Otherwise, it is normal. If $\eta_i > 1$, then that good is a luxury. It is often convenient to think of two goods: peanut butter and everything else; or consumption today and everything else.

At the initial allocation, $h_i = c_i$. Defining the expenditure share $s_j = p_j c_j / W$, we obtain

$$\boxed{\varepsilon_{ij}^M = \varepsilon_{ij}^H - s_j \eta_i.} \tag{3}$$

Thus, the Marshallian price elasticity equals the Hicksian price elasticity minus the expenditure share of good j multiplied by the wealth elasticity of good i. Notice that we are using cross-price elasticities here, but it is possible that $i = j$. The Marshallian own-price elasticity of demand for peanut butter is its Hicksian elasticity minus the share of your income that you spend on peanut butter times your income elasticity of demand for it!

3 Intertemporal Consumption (5 minutes)

Consider a two-period consumer with preferences $U(c_0, c_1)$ and budget constraint

$$c_0 + \frac{c_1}{R} = W, \qquad W = y_0 + \frac{y_1}{R}. \tag{4}$$

The price of future consumption is $p_1 = 1/R$. The ordinary Slutsky equation for current consumption is

$$\boxed{\frac{\partial c_0}{\partial p_1} = \frac{\partial h_0}{\partial p_1} - c_1 \frac{\partial c_0}{\partial W}.} \tag{5}$$

In an endowment economy, $W = y_0 + p_1 y_1$ also changes with p_1. Applying the chain rule yields

$$\boxed{\frac{dc_0}{dp_1} = \frac{\partial h_0}{\partial p_1} + (y_1 - c_1) \frac{\partial c_0}{\partial W}.} \tag{6}$$

This distinction matters when interpreting an interest-rate change.

4 Three Preference Specifications (12 minutes)

Throughout, let $0 < \beta \leq 1$ denote the discount factor.

4.1 Perfect Substitutes

Suppose

$$U(c_0, c_1) = c_0 + \beta c_1.$$

The consumer compares the payoff from current consumption with the discounted payoff from future consumption. Marshallian demands are

$$(c_0, c_1) = \begin{cases} (W, 0), & \beta R < 1, \\ (0, RW), & \beta R > 1. \end{cases} \tag{7}$$

At $\beta R = 1$, every affordable bundle is optimal. Hicksian demands are

$$(h_0, h_1) = \begin{cases} (\bar{u}, 0), & \beta R < 1, \\ (0, \bar{u}/\beta), & \beta R > 1. \end{cases} \tag{8}$$

At the threshold, all utility-minimizing-cost bundles satisfying $h_0 + \beta h_1 = \bar{u}$ are optimal. **Lesson:** substitution can be extreme, and ordinary price derivatives and elasticities need not exist at the threshold.

4.2 Perfect Complements

Suppose

$$U(c_0, c_1) = \min\{c_0, \beta c_1\}.$$

The consumer chooses $c_0 = \beta c_1$. Substituting into the budget constraint gives Marshallian demands

$$\boxed{c_0 = \frac{\beta RW}{1 + \beta R}, \qquad c_1 = \frac{RW}{1 + \beta R}.} \tag{9}$$

Hicksian demands are

$$\boxed{h_0 = \bar{u}, \qquad h_1 = \frac{\bar{u}}{\beta}.} \tag{10}$$

Lesson: the Hicksian substitution effect is zero; the price response at fixed wealth is entirely an income effect.

4.3 Log-Linear Preferences

Suppose

$$U(c_0, c_1) = \ln c_0 + \beta \ln c_1.$$

The first-order conditions are

$$\frac{1}{c_0} = \lambda, \qquad \frac{\beta}{c_1} = \frac{\lambda}{R},$$

so $c_1 = \beta R c_0$. The budget constraint then gives

$$\boxed{c_0 = \frac{W}{1 + \beta}, \qquad c_1 = \frac{\beta RW}{1 + \beta}.} \tag{11}$$

For current consumption, $\eta_0 = 1$ and the future-consumption expenditure share is $s_1 = \beta/(1 + \beta)$. Holding wealth fixed, c_0 does not depend on p_1, so $\varepsilon_{01}^M = 0$. The elasticity form of Slutsky gives

$$\boxed{0 = \underbrace{\frac{\beta}{1 + \beta}}_{\varepsilon_{01}^H} - \underbrace{\frac{\beta}{1 + \beta}}_{s_1 \eta_0}.} \tag{12}$$

Lesson: the substitution effect and the demand-side income effect cancel exactly for current consumption.

5 Conclusion: Two Income Channels (3 minutes)

Return to the endowment economy, where

$$W = y_0 + \frac{y_1}{R}.$$

An increase in R operates through two income channels. First, it reduces the present value of future endowment income:

$$\frac{dW}{dR} = -\frac{y_1}{R^2} < 0 \qquad (y_1 > 0).$$

Second, it produces the ordinary demand-side income effect in the Slutsky equation. For log-linear preferences, that second effect exactly offsets the substitution effect for current consumption. Since $c_0 = W/(1+\beta)$,

$$\boxed{\frac{dc_0}{dR} = -\frac{y_1}{(1+\beta)R^2}.} \tag{13}$$

Central conclusion: For log-linear preferences, the entire response of current consumption to an interest-rate change comes from the change in the present value of endowment income.